

Retroperitoneal lymph node dissection (RPLND)

If you have testicular cancer you may occasionally need surgery called retroperitoneal lymph node dissection (RPLND). This surgery is done to remove the lymph nodes at the back of the tummy (abdomen).

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What is retroperitoneal lymph node dissection (RPLND)?

[Retroperitoneal lymph nodes](#) are the nodes at the back of your tummy (abdomen). Sometimes these nodes need to be removed after having chemotherapy. This is called a retroperitoneal lymph node dissection (RPLND).

The lymph nodes may shrink after having [chemotherapy](#). But they may not disappear completely. After chemotherapy you have a [CT scan](#). If this shows signs the lymph nodes might still contain cancer cells, you usually need an RPLND. This is because removing these lymph nodes is the only way to know for sure.

Rarely, these lymph nodes are removed with an [early-stage non-seminoma](#) when someone cannot have [monitoring \(surveillance\)](#) or [adjuvant chemotherapy](#). An RPLND shows whether the cancer has spread and removes any nodes that might be affected.

How is the operation done?

Having an RPLND is a major operation. It is only done by experienced surgeons in specialist centres. Your surgeon and specialist nurse will explain what is involved, and the possible complications and side effects.

You have the operation done under a general anaesthetic. The surgeon makes a long cut from the top of your tummy (near your breastbone) to below your belly button.

The lymph nodes are in front of the spine. This means your surgeon has to move your bowel and other organs aside to reach them.

Keyhole surgery for RPLND

Sometimes, surgeons can do this operation using [keyhole surgery](#). For example, this may be an option if you have not previously had chemotherapy. This means you will have several small cuts (incisions) rather than one longer one. This is sometimes called laparoscopic or robotic retroperitoneal lymph node dissection. Keyhole surgery is very specialised, and only experienced surgeons should do it. There are only a few hospitals that can offer this type of surgery.

As keyhole surgery involves several small cuts, it can be quicker to recover from than standard surgery, where you have one longer cut.

Before your operation

If you smoke, try to [give up or smoke less](#) before your operation. This will help reduce your risk of chest problems, such as a chest infection. It will also help your wound to heal after the operation. Your GP can give you advice. You can also usually self-refer to a local stop-smoking service. The [NHS](#) has a lot of information and support to help you give up smoking. Look on the NHS website for the country where you live.

You will meet a member of the surgical team and a specialist nurse to discuss the operation. This may be at a pre-operative assessment clinic before you are admitted for your operation. You may also meet the doctor who will give you your anaesthetic (anaesthetist). It is important to tell them if you have had [bleomycin](#) chemotherapy as a treatment.

Make sure you ask any questions and discuss any concerns you have about the operation. If you need help when you go home after surgery, tell your hospital team as soon as possible. For example, you may be a carer for someone else or live alone. This gives your team plenty of time to make arrangements to help you.

You may have tests before the operation, such as:

- a [chest x-ray](#).
- [blood tests](#)
- a urine test
- an ECG (a recording of your heart).

You will usually be admitted to hospital on the morning of the operation. You will then meet members of the surgical team and nursing team, including your anaesthetist. They will be able to answer any questions you have about the anaesthetic and about pain control after the operation.

Recovering from an RPLND operation

You may be looked after in a high-dependency or intensive care unit for the first few days. You will be monitored very closely after your operation.

You will have:

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fluids directly into a vein by an infusion (drip) until you can eat and drink normally

- a tube draining your bladder (catheter), attached to a bag to collect urine
- a tube draining from your wound, attached to a bottle to collect excess fluid and help the wound to heal.

You will only need these tubes for a short time. They will usually be taken out before you go home.

Your treatment team will encourage you to start moving about as soon as possible. This helps to reduce complications like chest infections or blood clots. You will also need to wear support stockings (TED stockings). These help prevent blood clots in your legs. A physiotherapist or specialist nurse may give you some gentle leg and breathing exercises to do.

If you have pain, let your nurse or doctor know. You will have painkillers regularly to control any pain. But if you still have pain, you can have a different painkiller or a higher dose.

Possible complications of RPLND surgery

All operations have possible complications and these will depend on the type of operation you have. Your surgeon will discuss these with you before your operation.

The most common complications after surgery are:

- a wound infection
- bleeding in the operation area
- a chest infection
- developing a blood clot.

The nurses will monitor you for these complications. Let them know straight away if you have:

- any bleeding, or feel unwell
- symptoms of an infection, such as a cough or feeling short of breath
- discharge from your wound
- swelling and redness in a limb.

With RPLND surgery, the bowel may also take a while to start working normally again. Your doctors and nurses will do regular checks to look for signs of complications. This means that if any develop, they can treat them straight away.

Sometimes during RPLND surgery, the lymph nodes are found attached to a nearby structure, such as a kidney. If the surgeon cannot separate the lymph nodes from the kidney, they may need to remove both. But this is very rare.

Effects on fertility

A common side effect of this operation is dry ejaculation. You will still be able to ejaculate, but your semen will go into your bladder instead. This is known as retrograde ejaculation or dry climax. The semen will then leave your body when you next pass urine (pee).

If you cannot ejaculate in the usual way, you will not be able to get someone pregnant without fertility treatment. But you should still be able to get an erection and have an orgasm.

This side effect is due to nerves being damaged during the operation. It is more likely if you have previously had radiotherapy to the lymph nodes at the back of the tummy. This can cause some scarring of the nerves. Sometimes, surgeons can adapt the surgery to use nerve-sparing techniques that try to protect the nerves.

Your doctor will advise you to [store sperm](#) before the operation, if this is appropriate for you.

Going home

How long you need to be in hospital depends on how quickly you recover and whether you have any complications. It may take up to 10 days before you are ready to go home. If necessary, a district nurse can change your wound dressings at home. Most surgeons use dissolvable stitches, which they do not need to remove. You will have a long, vertical wound, which will be red and swollen at first. This will heal, but will leave a long scar that will gradually fade.

It may take a few months to fully recover from your operation. Your doctor will advise when you can lift heavy objects (such as bags of shopping), drive and return to work.

For more support, visit our [testicular cancer forum](#) to talk with people who have been affected by testicular cancer, share your experience, and ask an expert your questions.

About our information

References



Below is a sample of the sources used in our testicular cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

The British Association of Urological Surgeons (BAUS) Guidelines on Testicular Cancer. March 2015. Available from: https://www.baus.org.uk/professionals/sections/testicular_cancer.aspx (accessed April 2022).

European Association of Urology (EAU) Guidelines on Testicular Cancer 2022. Available from: <https://d56bochluxqnz.cloudfront.net/documents/full-guideline/EAU-Guidelines-on-Testicular-Cancer-2022.pdf> (accessed April 2022).

European Society for Medical Oncology (ESMO). Testicular seminoma and non-seminoma: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Last updated 2022.

European Society for Medical Oncology (ESMO). Testicular seminoma and non-seminoma: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Last updated 2022. Available from: [https://www.annalsofoncology.org/article/S0923-7534\(22\)00007-2/fulltext](https://www.annalsofoncology.org/article/S0923-7534(22)00007-2/fulltext) (accessed April 2022).

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Ursula McGovern](#), Consultant Medical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

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⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

[Get help](#)